

CHESAPEAKE MONEY POINT, VA, USA

WMO: 997305

Lat: 36.778N

Lon: 76.302W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.7	(c) -3.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 9.7	(k) 4.2	(l) 8.4	(m) 6.4	(n) 3.2	(o) 0	(p) 0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.7	(c) 33.8	(d) N/A	(e) 32.6	(f) N/A	(g) 31.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.2	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.9	(b) 7.4	(c) 6.2	(e) -7.9	(f) 35.9	(g) 2.6	(h) 1.5	(i) -9.7	(j) 37.0	(k) -11.3	(l) 37.9	(m) -12.7	(n) 38.7	(o) -14.6	(p) 39.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec				
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)				
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.6	6.7	10.0	16.1	20.4	24.9	26.9	26.3	23.2	17.9	11.8	8.8	(6)		
(7)		DBStd	8.56	5.59	5.25	5.18	4.41	3.78	2.82	1.97	1.88	2.72	4.09	4.30	5.15	(7)		
(8)		HDD10.0	464	158	119	67	4	0	0	0	0	0	2	29	84	(8)		
(9)		HDD18.3	1655	394	328	262	93	23	1	0	1	57	199	297	(9)			
(10)		CDD10.0	2878	22	26	67	188	323	448	524	505	397	247	83	48	(10)		
(11)		CDD18.3	1028	0	1	4	27	88	198	266	247	148	44	3	3	(11)		
(12)		CDH23.3	8473	0	3	31	204	599	1719	2613	2176	912	204	5	6	(12)		
(13)		CDH26.7	2809	0	0	2	41	157	624	996	727	227	35	0	0	(13)		
(14)	Wind	WSAvg	2.8	2.7	3.0	3.4	3.2	3.1	2.7	2.6	2.3	2.3	2.6	2.7	2.6	(14)		
(15)	Precipitation	PrecAvg	1207	82	76	93	83	90	107	139	135	137	100	80	84	(15)		
(16)		PrecMax	1534	149	218	251	153	135	263	240	345	444	207	280	172	(16)		
(17)		PrecMin	782	28	24	16	15	34	42	33	40	17	7	4	34	(17)		
(18)		PrecStd	205	32	41	44	34	26	50	50	74	98	53	59	36	(18)		
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.6	24.0	26.5	30.1	32.5	35.0	35.7	34.5	32.5	29.7	24.2	24.1	(19)		
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
(21)		2%	DB	19.8	21.0	23.6	27.8	30.1	33.2	33.9	32.9	30.7	27.4	22.3	21.7	(21)		
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
(23)		5%	DB	17.4	19.0	21.1	25.8	28.6	31.7	32.6	31.6	29.3	25.6	20.5	19.7	(23)		
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
(25)		10%	DB	15.0	15.5	18.8	23.7	26.8	30.2	31.3	30.4	27.8	24.0	18.8	17.5	(25)		
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)		
(28)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
(30)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
(32)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
(34)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35)	Mean Daily Temperature Range			MDBR	8.1	8.5	8.7	9.4	8.5	7.9	7.7	7.0	6.7	7.7	7.9	8.0	(35)	
(36)		5% DB			MCD BR	11.5	11.9	12.5	12.2	10.3	10.1	9.5	8.9	8.7	9.3	10.1	9.9	(36)
(37)					MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)			5% WB			MCD BR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(39)				MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.309	0.322	0.348	0.389	0.433	0.483	0.508	0.501	0.404	0.357	0.325	0.313	(40)		
(41)		taud		2.535	2.494	2.449	2.347	2.255	2.169	2.113	2.118	2.381	2.505	2.543	2.553	(41)		
(42)		Ebn at Noon		889	917	918	893	854	809	785	781	848	865	861	859	(42)		
(43)		Edh at Noon		78	92	105	123	137	149	157	153	112	90	77	73	(43)		
(44)	All-Sky Solar Radiation	RadAvg		2.36	3.12	4.16	5.30	5.96	6.37	6.07	5.47	4.46	3.56	2.67	2.11	(44)		
(45)		RadStd		0.19	0.33	0.32	0.39	0.50	0.40	0.30	0.32	0.35	0.41	0.21	0.17	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+105

Nomenclature: See separate page